

13. Pembroke-Graham's Pond

Shoreline length: 12 km



Shore Types

Pembroke-Graham's Pond is an exposed coast along the east end of PEI, with a majority of wetlands, low plains, and dunes.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
18	2	2	12	41	5	17	3

Governance, Infrastructure, and Economy

Pembroke-Graham's Pond SMU includes a portion within the Town of Three Rivers (~19%), with the remaining shoreline being entirely unincorporated and subject to provincial planning authority.

Coastal infrastructure includes the Graham's Pond Harbour, along with Graham's Pond Road (Gaspereaux), Route 17 at Graham's Pond, and MacLure Pond Road (Gaspereaux).

Coastal economic drivers include both Gaspereaux Harbour and the Graham's Pond wharf support active fisheries. Local and legacy land use constraints include a few small coastal lots and unpaved coastal roads.



Graham's Pond Harbour

Natural Assets and Heritage

Popular Poverty Beach is located at the south end of the SMU

The shoreline includes critical habitat for bird species including the endangered Piping Plover.



Irving's Cape (left), Poverty Beach and down-drift sandspit (right) at south end of SMU

Planning and Policy Framework

Where provincial rules apply, land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are

typically conducted. Within Three Rivers, the Town's municipal rules apply.

Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Sediment transport

The net sediment transport across the SMU is high and flows southward towards Poverty Beach and its terminal sandspit, which provides partial shelter to Murray Harbour North.

Coastal Flood and Erosion Risk

While low plains, dunes and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. Erosion risk is low due to the sparse amount of infrastructure and buildings. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline] - Note: Grahams Pond Harbour includes most of the infrastructure at-risk.

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	45 [2/km]	47 [2/km]	48 [2/km]	54 [3/km]	2 [<1/km]	4 [<1/km]	7 [<1/km]
Roads	1203 m [57 m/km]	1419 m [67 m/km]	1708 m [81 m/km]	2059 m [97 m/km]	62 m [3 m/km]	162 m [8 m/km]	307 m [15 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
13.00 Pembroke-Grahams Pond	NAI	MR	MR
Rationale	The initial strategy proposed is No Active Intervention, which will preserve natural shorelines and sediment supply to beaches. Managed realignment is proposed in the future which will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including dunes, wetlands and cliff habitat.		
13.01 Grahams Pond Harbour -PU	R	R	R
Rationale	Resist strategy would be suitable for the harbour as key infrastructure to the community. Future works would also require flood mitigation, subject to final strategies to be developed with the local authority.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.